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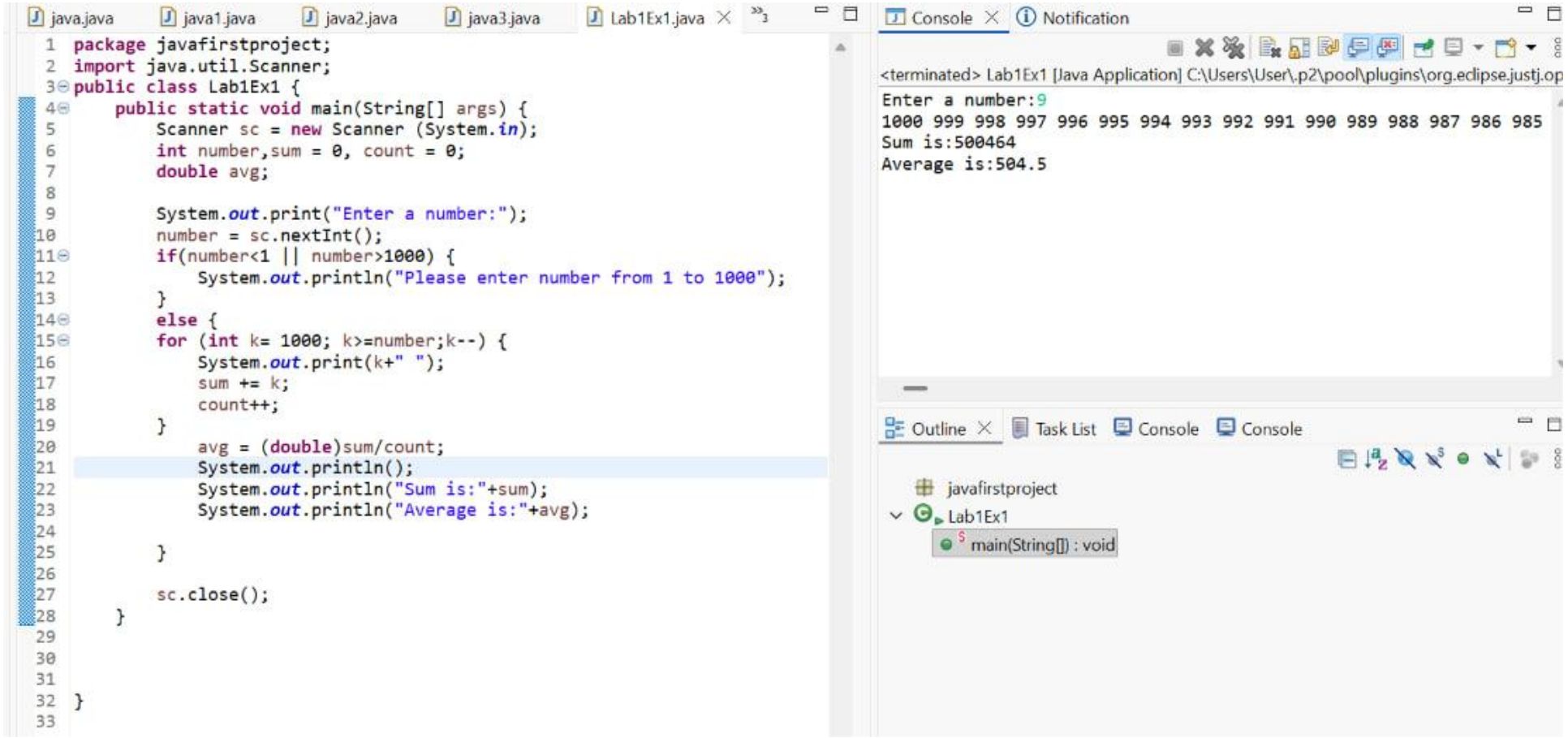
Lab01: Basic Java syntax

I2-IP-group

Lab1-Ex1: ……..

……(add instruction here ……..

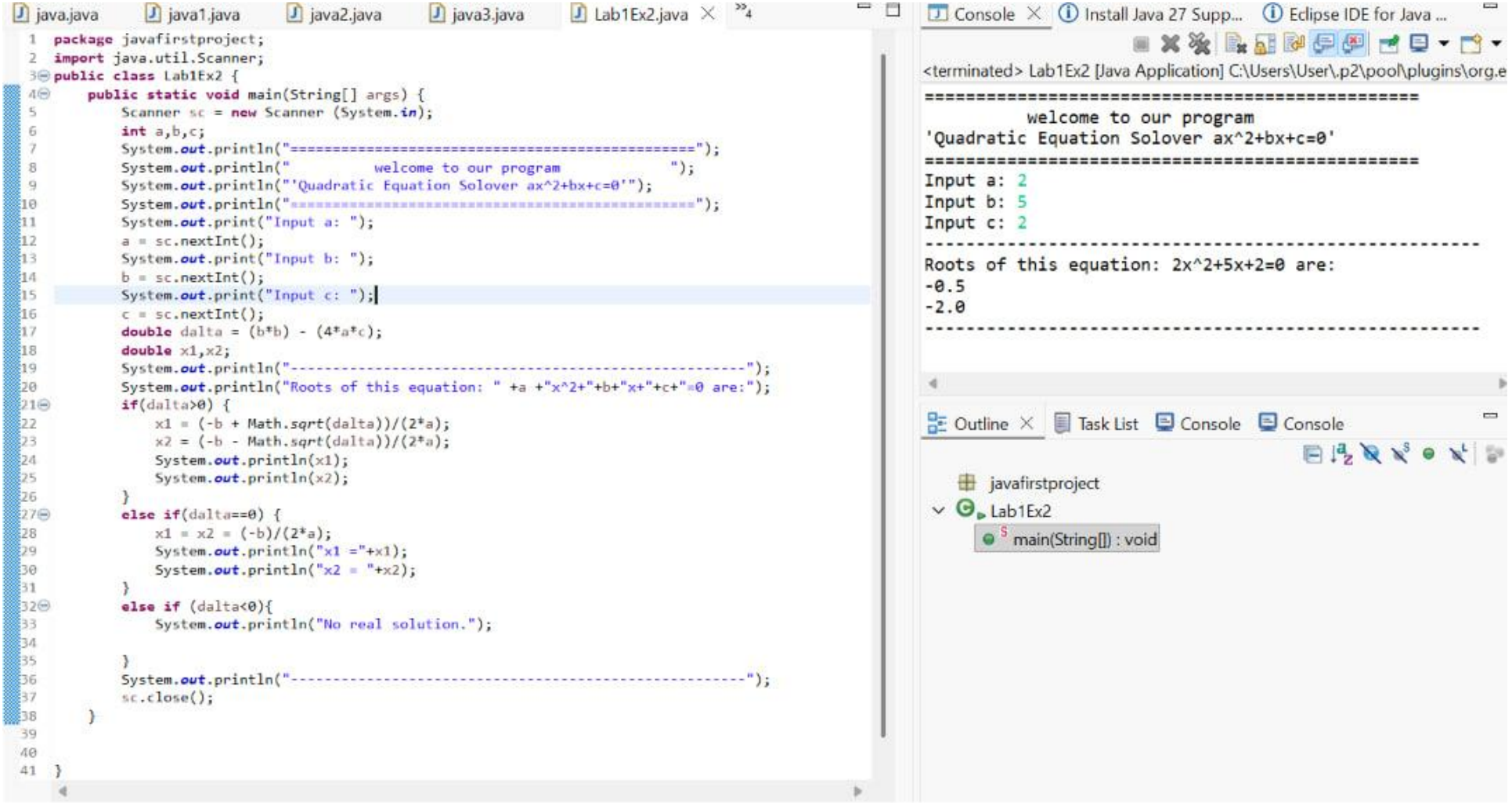
SOLUTION :



Lab1-Ex2: ……..

……(add instruction here ……..

SOLUTION:



```
package javafirstproject;
import java.util.Scanner;
public class Lab1Ex2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner (System.in);
        int a,b,c;
        System.out.println("================================================");
        System.out.println("          welcome to our program              ");
        System.out.println("'Quadratic Equation Solover ax^2+bx+c=0'");
        System.out.println("================================================");
        System.out.print("Input a: ");
        a = sc.nextInt();
        System.out.print("Input b: ");
        b = sc.nextInt();
        System.out.print("Input c: ");
        c = sc.nextInt();
        double dalta = (b*b) - (4*a*c);
        double x1,x2;
        System.out.println("------------------------------------------------------");
        System.out.println("Roots of this equation: " +a +"x^2+"+b+"x+"+c+"=0 are:");
        if(dalta>0) {
            x1 = (-b + Math.sqrt(dalta))/(2*a);
            x2 = (-b - Math.sqrt(dalta))/(2*a);
            System.out.println(x1);
            System.out.println(x2);
        }
        else if(dalta==0) {
            x1 = x2 = (-b)/(2*a);
            System.out.println("x1 ="+x1);
            System.out.println("x2 = "+x2);
        }
        else if (dalta<0){
            System.out.println("No real solution.");

        }
        System.out.println("------------------------------------------------------");
        sc.close();
    }

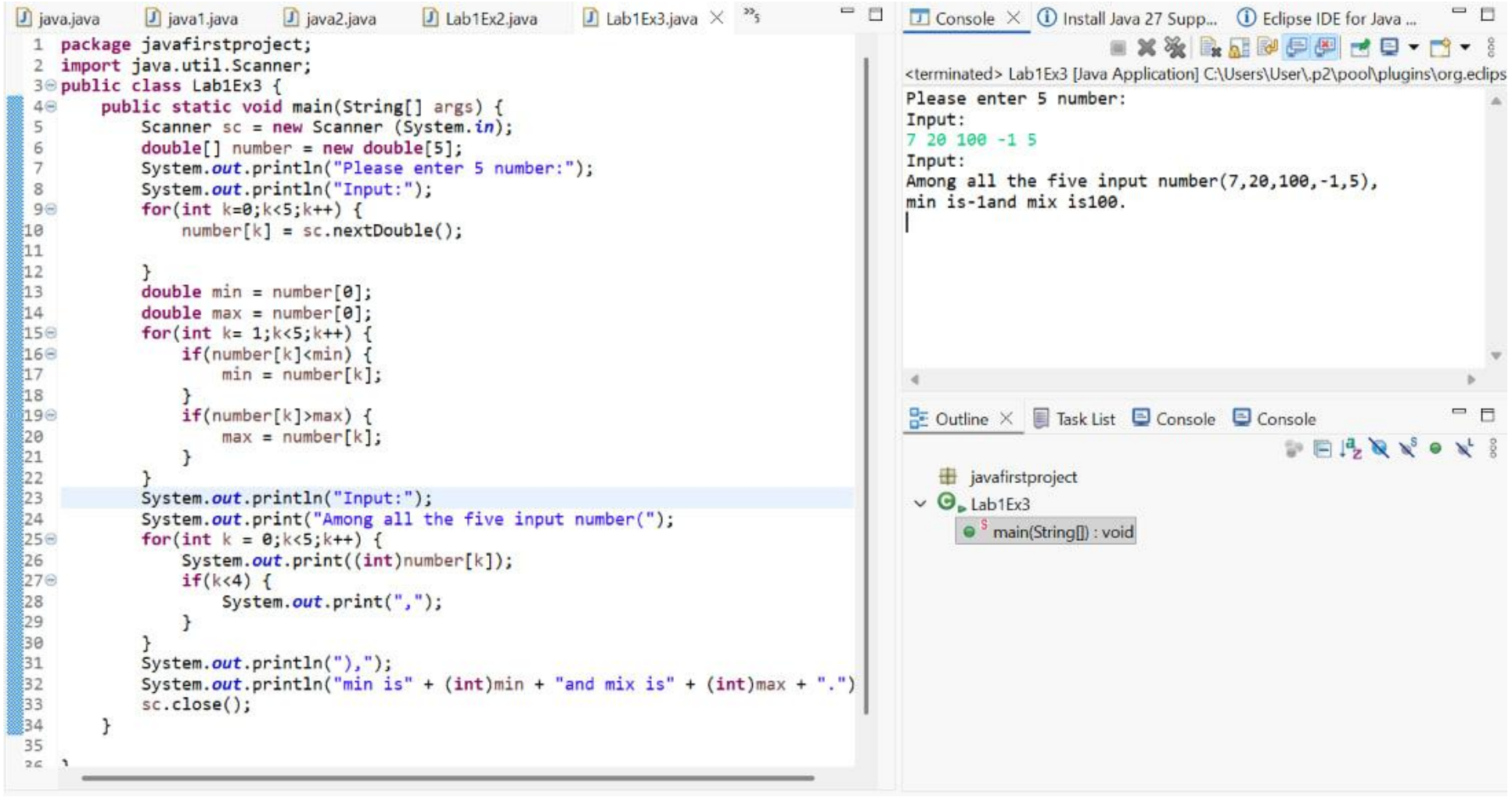

}
```

```
<terminated> Lab1Ex2 [Java Application] C:\Users\User\.p2\pool\plugins\org.e
================================================
          welcome to our program
'Quadratic Equation Solover ax^2+bx+c=0'
================================================
Input a: 2
Input b: 5
Input c: 2
------------------------------------------------------
Roots of this equation: 2x^2+5x+2=0 are:
-0.5
-2.0
------------------------------------------------------
```

Lab1-Ex3: ……..

……(add instruction here ……..

SOLUTION



```
package javafirstproject;
import java.util.Scanner;
public class Lab1Ex3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner (System.in);
        double[] number = new double[5];
        System.out.println("Please enter 5 number:");
        System.out.println("Input:");
        for(int k=0;k<5;k++) {
            number[k] = sc.nextDouble();

        }
        double min = number[0];
        double max = number[0];
        for(int k= 1;k<5;k++) {
            if(number[k]<min) {
                min = number[k];
            }
            if(number[k]>max) {
                max = number[k];
            }
        }
        System.out.println("Input:");
        System.out.print("Among all the five input number(");
        for(int k = 0;k<5;k++) {
            System.out.print((int)number[k]);
            if(k<4) {
                System.out.print(",");
            }
        }
        System.out.println("),");
        System.out.println("min is" + (int)min + "and mix is" + (int)max + ".")
        sc.close();
    }

}
```

```
<terminated> Lab1Ex3 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclips
Please enter 5 number:
Input:
7 20 100 -1 5
Input:
Among all the five input number(7,20,100,-1,5),
min is-1and mix is100.
```

Outline

javafirstproject

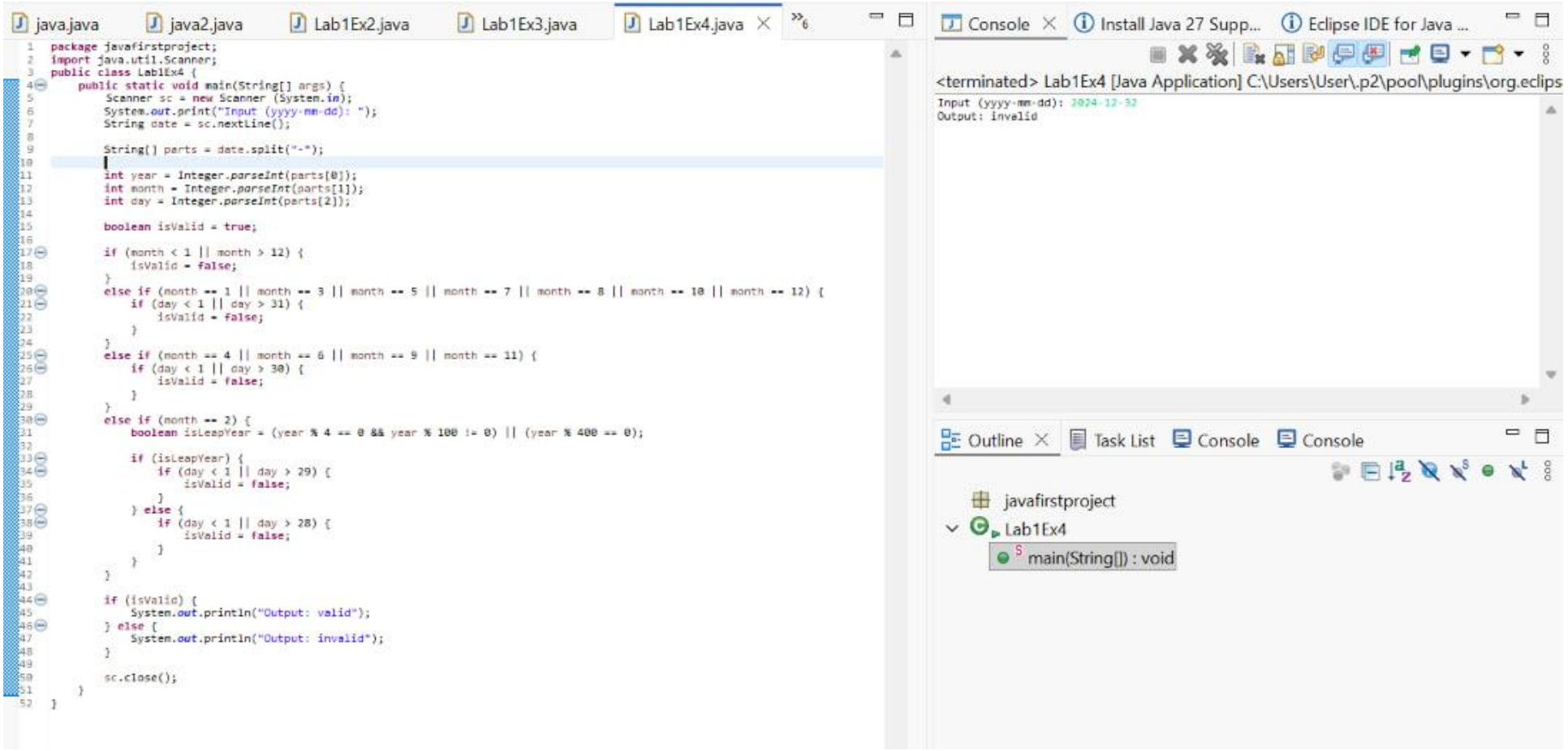
Lab1Ex3

main(String[]) : void

Lab1-Ex4: ……..

……(add instruction here ……..

SOLUTION




```java
package javafirstproject;
import java.util.Scanner;
public class Lab1Ex4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print("Input (yyyy-mm-dd): ");
        String date = sc.nextLine();

        String[] parts = date.split("-");

        int year = Integer.parseInt(parts[0]);
        int month = Integer.parseInt(parts[1]);
        int day = Integer.parseInt(parts[2]);

        boolean isValid = true;

        if (month < 1 || month > 12) {
            isValid = false;
        }
        else if (month == 1 || month == 3 || month == 5 || month == 7 || month == 8 || month == 10 || month == 12) {
            if (day < 1 || day > 31) {
                isValid = false;
            }
        }
        else if (month == 4 || month == 6 || month == 9 || month == 11) {
            if (day < 1 || day > 30) {
                isValid = false;
            }
        }
        else if (month == 2) {
            boolean isLeapYear = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);

            if (isLeapYear) {
                if (day < 1 || day > 29) {
                    isValid = false;
                }
            } else {
                if (day < 1 || day > 28) {
                    isValid = false;
                }
            }
        }

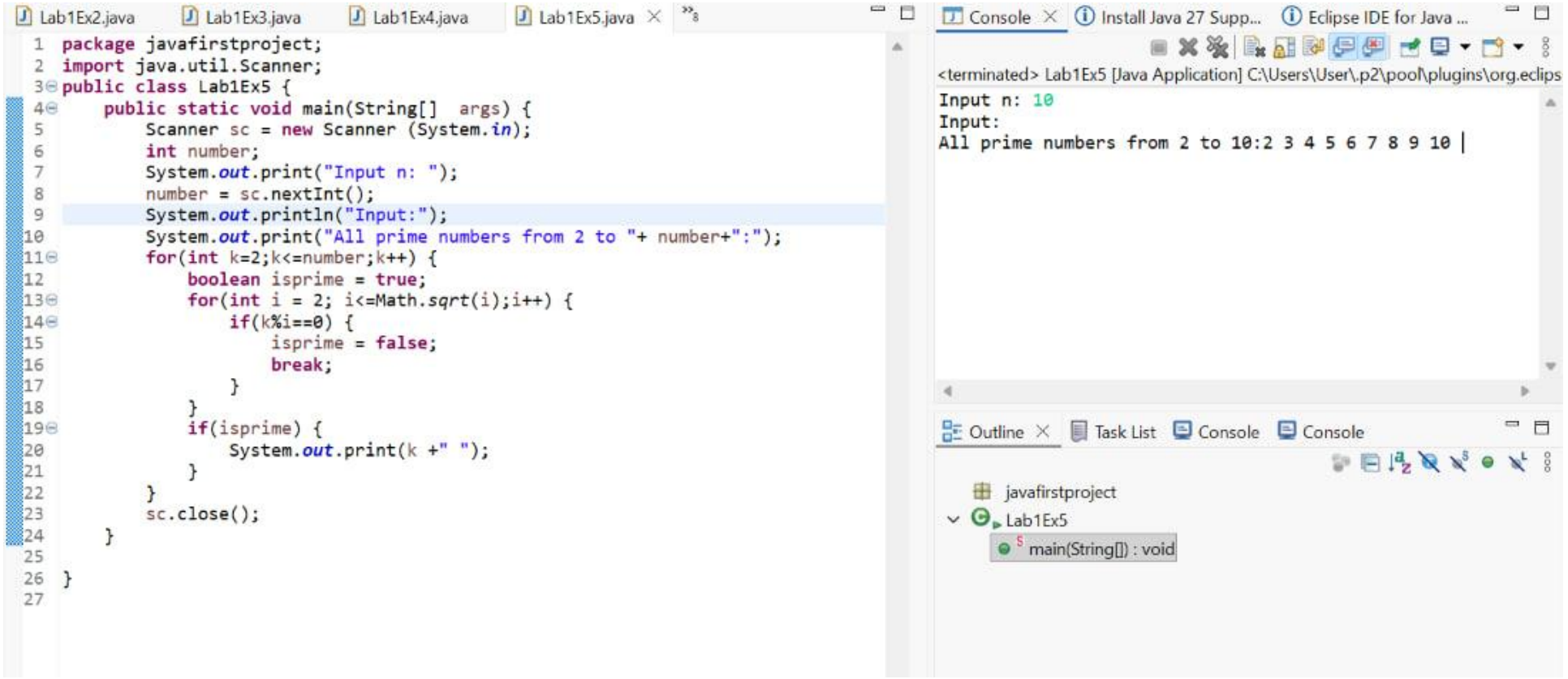
        if (isValid) {
            System.out.println("Output: valid");
        } else {
            System.out.println("Output: invalid");
        }

        sc.close();
    }
}
```

Lab1-Ex5: ……..

……(add instruction here ……..

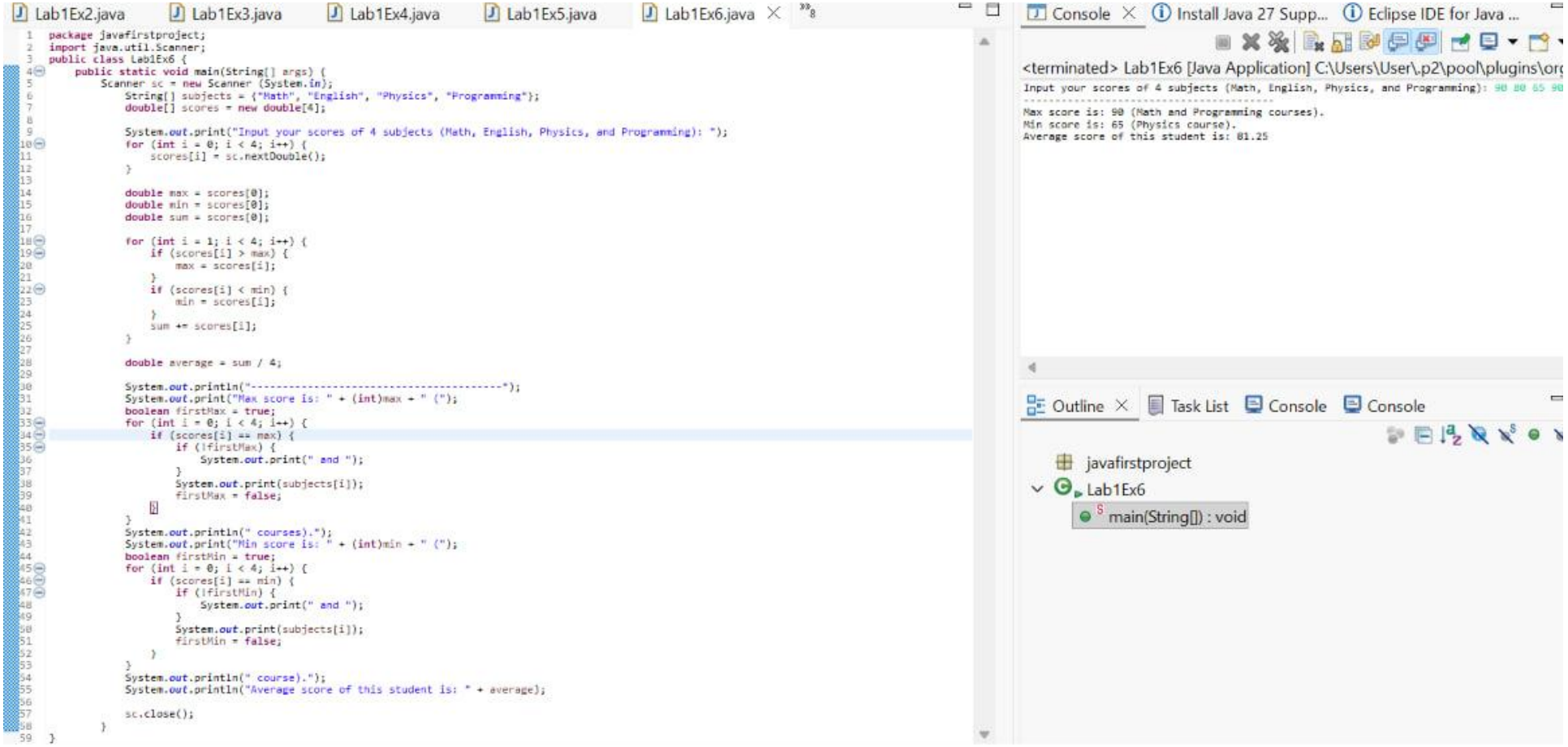
SOLUTION



Lab1-Ex6: ……..

……(add instruction here ……..

SOLUTION



```
Lab1Ex2.java  Lab1Ex3.java  Lab1Ex4.java  Lab1Ex5.java  Lab1Ex6.java

1  package javafirstproject;
2  import java.util.Scanner;
3  public class Lab1Ex6 {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner (System.in);
6              String[] subjects = {"Math", "English", "Physics", "Programming"};
7              double[] scores = new double[4];
8
9              System.out.print("Input your scores of 4 subjects (Math, English, Physics, and Programming): ");
10             for (int i = 0; i < 4; i++) {
11                 scores[i] = sc.nextDouble();
12             }
13
14             double max = scores[0];
15             double min = scores[0];
16             double sum = scores[0];
17
18             for (int i = 1; i < 4; i++) {
19                 if (scores[i] > max) {
20                     max = scores[i];
21                 }
22                 if (scores[i] < min) {
23                     min = scores[i];
24                 }
25                 sum += scores[i];
26             }
27
28             double average = sum / 4;
29
30             System.out.println("----------------------------------------");
31             System.out.print("Max score is: " + (int)max + " (");
32             boolean firstMax = true;
33             for (int i = 0; i < 4; i++) {
34                 if (scores[i] == max) {
35                     if (!firstMax) {
36                         System.out.print(" and ");
37                     }
38                     System.out.print(subjects[i]);
39                     firstMax = false;
40                 }
41             }
42             System.out.println(" courses).");
43             System.out.print("Min score is: " + (int)min + " (");
44             boolean firstMin = true;
45             for (int i = 0; i < 4; i++) {
46                 if (scores[i] == min) {
47                     if (!firstMin) {
48                         System.out.print(" and ");
49                     }
50                     System.out.print(subjects[i]);
51                     firstMin = false;
52                 }
53             }
54             System.out.println(" course).");
55             System.out.println("Average score of this student is: " + average);
56
57             sc.close();
58         }
59 }
```

```
Console  Install Java 27 Supp...  Eclipse IDE for Java ...

<terminated> Lab1Ex6 [Java Application] C:\Users\User\.p2\pool\plugins\or
Input your scores of 4 subjects (Math, English, Physics, and Programming): 90 80 65 90
----------------------------------------
Max score is: 90 (Math and Programming courses).
Min score is: 65 (Physics course).
Average score of this student is: 81.25
```

```
Outline  Task List  Console  Console

javafirstproject
Lab1Ex6
main(String[]) : void
```